higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

ELECTRONIC CONTROL AND DIGITAL ELECTRONICS
NQF LEVEL 3

(12041023)

14 November 2019 (X-Paper)
09:00–12:00

This question paper consists of 6 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each question on a NEW page.
5. Write neatly and legibly.

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QUESTION 1: FUNDAMENTALS OF ELECTRONICS

1.1 1.1.1 Function generators are used by electronic technicians.

State TWO uses of this test instrument. (2)

1.1.2 As a technician, you suspect that a resistor is faulty.

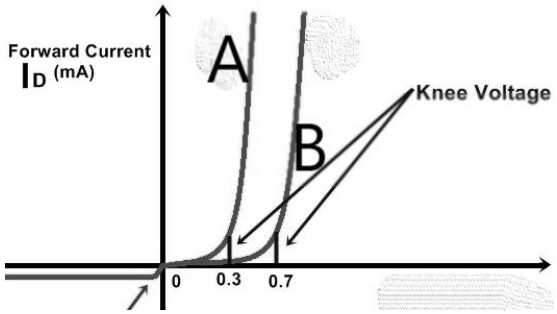
Explain how you would use a digital multimeter to test a resistor. (3)

1.2 1.2.1 Explain how atom energy bands and energy gaps (conduction and valence bands) change the characteristics of an insulator and conductor. (4)

1.2.2 Which type of bonding is formed by the mutual sharing of electrons? (1)

1.3 1.3.1 The characteristic curves of diodes are represented by Graph 1 below.

Name the types of semiconductor material that represent the curve:



GRAPH 1

(a) A

(b) B

(2 × 1) (2)

1.3.2 Explain how the biasing of a tunnel diode allows it to operate as a variable capacitor. (3)

1.3.3 The silicon diode specification sheets have symbols that describe its electrical characteristics. What do the following symbols indicate:

(a) V_F

(b) I_R

(2 × 1) (2)

1.4 1.4.1 A switching transistor has biasing currents of I_E = 10 mA and I_B = 3 mA.

Calculate its value of I_C. (2)

1.4.2 Draw the non-biased common emitter circuit diagram configuration, indicating input and output waveforms. (4)

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1.5 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.5.1–1.5.5) in the ANSWER BOOK.

1.5.1 Centre-tapped transformers are cheaper than basic transformers.

1.5.2 The output of the transformer produces half wave pulses of DC.

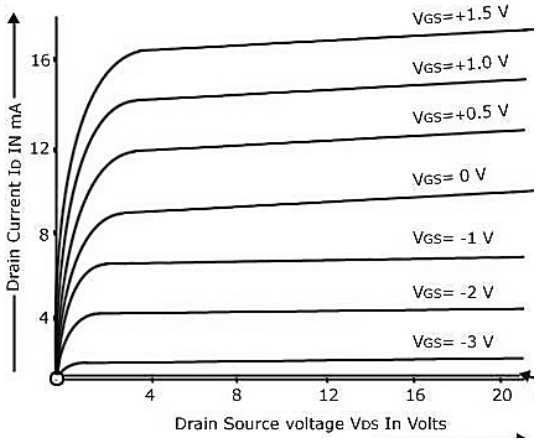
1.5.3 Each half cycle flows through one diode only.

1.5.4 Electrolytic capacitors can be used to filter the output DC voltage.

1.5.5 An LM7809 IC can be used in this circuit to regulate a negative output voltage.

(5 × 1) (5)

1.6 1.6.1 Analyse the characteristic curve (Graph 2) of an N-channel depletion type MOSFET and explain its operation.



GRAPH 2 (4)

1.6.2 How does Electrostatic Discharge (ESD) damage a MOSFET when not protected? (2)

1.7 1.7.1 An inverting Op-amp has the following resistors biasing the IC: R_F = 180 kΩ and R_{IN} = 30 kΩ. The input voltage V_{IN} = 0,5 V.

Calculate the output amplitude – V_{OUT}. (3)

1.7.2 What are the following Op-amp circuits used for (application)?

(a) Integrator

(b) Voltage follower

(c) Audio amplifier

(3 × 1) (3)

[40]

QUESTION 2: DIGITAL ELECTRONICS

2.1 Convert between the following number systems:

2.1.1 11011_2 to a decimal number

2.1.2 45_8 to a binary number

2.1.3 236_{10} to a hexadecimal number

(3 × 3) (9)

2.2 Binary arithmetic

2.2.1 Subtract the following binary numbers:

$100010_2 - 11101_2$

2.2.2 Multiply the following binary numbers:

$10110_2 \times 1001_2$

2.2.3 Divide the following binary numbers:

$11101_2 \div 110_2$

(3 × 3) (9)

2.3 2.3.1 Describe the operation of an asynchronous decade counter. (3)

2.3.2 Design a block circuit diagram of a 2 bit synchronous up counter. (4)

[25]

QUESTION 3: ASSEMBLING A PERSONAL COMPUTER

3.1 3.1.1 You have assembled a client's desktop computer, but it does not want to start up.

What steps would you take to troubleshoot this fault? (3)

3.2 Explain what is meant by the word *cache*. (2)

3.3 What are the following ports on the mother board of a computer used for?

3.3.1 VGA

3.3.2 PCI

(2 × 1) (2)

3.4 A desktop tower needs to have its CPU IC replaced.

Explain how you would remove and replace this sensitive component, keeping ESD safety in mind also. (8)

[15]

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QUESTION 4: TRANSDUCERS

4.1 In the field of instrumentation, explain what are transducers used for. (2)

4.2 A photo sensitive transducer uses light to operate.

Explain how this device can be used to measure the speed of a rotating disc. (5)

4.3 A Linear Variable Differential Transformer (LVDT) transducer uses a modified transformer to operate.

Discuss the operating principle of this device. (3)

[10]

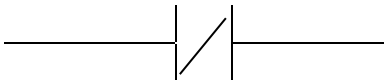
QUESTION 5: PROGRAMMABLE LOGIC CONTROLLERS

5.1 A laptop is an example of a programming device used by a PLC programmer.

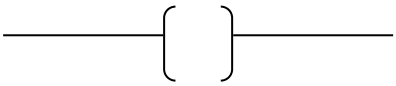
Identify TWO uses for this device in PLC operation. (2)

5.2 Identify and indicate the state of the following ladder logic symbols:

5.2.1



5.2.2



(2 × 1) (2)

5.3 A motor M is started by one push to make button A. After releasing the button, the motor continues to run. The operator can then stop the motor from two different locations: Stop buttons B and C (OR configuration).

Draw the ladder logic diagram for the operation above. (6)

[10]

TOTAL: 100

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